The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 Dec 93.

INCH-POUND

MIL-S-19500/354D
10 September 1993
SUPERSEDING
MIL-S-19500/354C
21 June 1983

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPES 2N2604 AND 2N2605, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, silicon, low-power transistors for use in low noise-level amplifier applications. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

Type	P_T 1/ T_A = +25°C	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_J and T_{STG}	$R_{\theta JA}$
	mW	V dc	V dc	V dc	mA dc	°C	°C/mW
2N2604	400	80	6	60	30	-65 to +200	.437
2N2605	400	70	6	60	30	-65 to +200	.437

1/ Derate linearly at 2.28 mW/°C above T_A = +25°C.

1.4 Primary electrical characteristics.

	h_{FE1}		h_{fe}		$\lvert h_{fe} \rvert$	C_{obo}	$V_{BE(sat)}$	$V_{CE(sat)}$
	V_{CE} = 5 V dc I_C = 10 μ dc		V_{CE} = 5 V dc I_C = 1 mA dc f = 1 kHz		V_{CE} = 5 V dc I_C = 500 μA dc f = 30 MHz	V_{CB} = 5 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	I_C = 10 mA dc I_B = 500 μA dc	I_C = 10 mA dc I_B = 500 μA dc
	2N2604	2N2605	2N2604	2N2605		pF	V dc	V dc
Min	40	100	60	150	1		0.7	
Max	120	300	180	450	8	6	0.9	0.3

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Research Laboratory, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ 07703-5601 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 9, T-2B.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CBO1} and h_{FE1}	Not applicable
11	I_{CBO1}; h_{FE1}; ΔI_{CBO1} = 100 percent or 2 nA dc, whichever is greater; Δh_{FE1} = ±25 percent change of initial value.	I_{CBO1} and h_{FE1}
12	See 4.2.1	See 4.2.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100 percent or 2 nA dc, whichever is greater; Δh_{FE1} = ±25 percent change of initial value.	Subgroup 2 of table I herein; ΔI_{CBO1} = 100 percent or 2 nA dc, whichever is greater; Δh_{FE1} = ±25 percent change of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

V_{CB} = 25 V dc; P_T = 400 mW at T_A = room ambient as defined in 4.5 of MIL-STD-750.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 20 V dc; P_T = 400 mW at T_A = room ambient as defined in 4.5 of MIL-STD-750; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 20 V dc; T_A = +125°C ±25°C for 96 hours, P_T = 400 mW at T_A = +100°C or adjusted as required by the chosen T_A to give an average lot. T_J = +275°C.

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4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	V_{CB} = 25 V dc; P_T = 400 mW at T_A = room ambient as defined in 4.5 of MIL-STD-750. No heat sink or forced-air cooling on the devices shall be permitted.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
C2	2036	Test condition E
C6	1026	V_{CB} = 25 V dc; P_T = 400 mW at T_A = room ambient as defined in 4.5 of MIL-STD-750. No heat sink or forced-air cooling on device shall be permitted.
C7	3131	$R_{\theta JA}$ = .437°C/mW. LTPD = 15.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Noise figure. The noise figure shall be measured using commercially available test equipment and its associated standard test procedures.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector - base breakdown voltage	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$			
2N2604				80		V dc
2N2605				70		V dc
Collector - emitter breakdown voltage	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc
Emitter - base breakdown voltage	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	6		V dc
Collector - base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO1}		10	nA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO}		2	nA dc
Collector - emitter cutoff current	3041	Bias condition C; V_{CE} = 50 V dc	I_{CES}		10	nA dc
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc	h_{FE1}			
2N2604				40	120	
2N2605				100	300	
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 500 µA dc	h_{FE2}			
2N2604				60	180	
2N2605				150	450	
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 mA dc	h_{FE3}			
2N2604				40	160	
2N2605				100	400	
Base - emitter voltage (saturated)	3066	Test condition A; I_C = 10 mA dc; I_B = 500 µA dc	$V_{BE(sat)}$	0.7	0.9	V dc
Collector - emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 500 µA dc	$V_{CE(sat)}$		0.3	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector - base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO2}		5	µA dc
Low-temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc	h_{FE4}			
2N2604				15		
2N2605				30		
Subgroup 4						
Small-signal short-circuit input impedance	3201	V_{CB} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{ie}			
2N2604				1	10	kΩ
2N2605				2	20	kΩ
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{re}		10 x	10^{-4}
Small-signal open-circuit output admittance	3216	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{oe}			
2N2604					40	µmhos
2N2605					60	µmhos
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}			
2N2604				60	180	
2N2605				150	450	
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 0.5 mA dc; f = 30 MHz	$\lvert h_{fe} \rvert$	1	8	
Open capacitance input open circuited	3236	V_{CB} = 5 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		6	pF
Noise figure	3246	V_{CE} = 5 V dc; I_C = 10 µA dc; R_g = 10 kΩ; f = 100 Hz	F_1		5	dB
Noise figure	3246	V_{CE} = 5 V dc; I_C = 10 µA dc; R_g = 10 kΩ; f = 1 kHz	F_2		3	dB
Noise figure	3246	V_{CE} = 5 V dc; I_C = 10 µA dc; R_g = 10 kΩ; f = 10 kHz	F_3		3	dB

1/ For sampling plan, see MIL-S-19500.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead formation and finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI
Navy - AS, CG, MC
Air Force - 13, 15, 19, 85
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/354D	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER, TYPES 2N2604 AND 2N2605, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, first, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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